

Chapter 11

Partial Differentiation

Why Should We Study Partial Differentiation?

Partial differentiation is used when a quantity depends on more than one variable.

This chapter helps us understand:

- Functions involving many variables
- Optimization problems in Economics and Engineering
- Heat flow and fluid mechanics
- Machine learning and Data Science
- Maximum profit and minimum cost problems
- Mathematical modelling of real-life systems

11.1 Introduction

Ordinary Differential Equations and Partial Differential Equations

Differential equations arise when derivatives of functions are involved. They are widely used in Mathematics, Physics, Engineering, Economics and Biology.

11.1.1 Ordinary Differential Equation (ODE)

Definition of ODE

A differential equation involving ordinary derivatives of a function with respect to a single independent variable is called an:

Ordinary Differential Equation (ODE)

General Form

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots\right) = 0$$

Examples

1.

$$\frac{dy}{dx} + y = 0$$

2.

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

3.

$$\frac{dy}{dx} = x + y$$

Explanation

In ODEs, the dependent variable depends on only one independent variable.

Example:

$$y = f(x)$$

11.1.2 Partial Differential Equation (PDE)**Definition of PDE**

A differential equation involving partial derivatives of a function with respect to two or more independent variables is called :

$$\boxed{\text{Partial Differential Equation (PDE)}}$$

General Form

$$F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \dots\right) = 0$$

Examples

1.

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$$

2.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

3.

$$\frac{\partial z}{\partial t} = k \frac{\partial^2 z}{\partial x^2}$$

Explanation

In PDEs, the dependent variable depends on two or more independent variables.

Example:

$$z = f(x, y)$$

11.2 Difference Between ODE and PDE

Feature	ODE	PDE
Derivatives Used	Ordinary Derivatives	Partial Derivatives
Independent Variables	One	Two or More
General Form	$\frac{dy}{dx}$	$\frac{\partial z}{\partial x}$
Example	$\frac{dy}{dx} + y = 0$	$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$

Important Note

- ODE involves derivatives with respect to one variable.
- PDE involves partial derivatives with respect to multiple variables.
- Differential equations are essential for modelling real-life systems.
- ODEs are commonly used in population growth and motion problems.
- PDEs are widely used in heat transfer, fluid dynamics and quantum mechanics.

11.3 Partial Derivatives

Suppose:

$$z = f(x, y)$$

Then:

$$\frac{\partial z}{\partial x}$$

means differentiation with respect to x , keeping y constant.

Similarly,

$$\frac{\partial z}{\partial y}$$

means differentiation with respect to y , keeping x constant.

Example

If:

$$z = x^2y + 3xy^2$$

Find:

$$\frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y}$$

Solution

Differentiating with respect to x :

$$\frac{\partial z}{\partial x} = 2xy + 3y^2$$

Differentiating with respect to y :

$$\frac{\partial z}{\partial y} = x^2 + 6xy$$

$$\frac{\partial z}{\partial x} = 2xy + 3y^2$$

$$\frac{\partial z}{\partial y} = x^2 + 6xy$$

11.4 Homogeneous Functions

A function is called homogeneous if all terms have the same degree.

Definition of Homogeneous Function

A function:

$$f(x, y)$$

is homogeneous of degree n if:

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

Example

Show that:

$$f(x, y) = x^2 + xy + y^2$$

is homogeneous.

Solution

Replace:

$$x \rightarrow \lambda x, \quad y \rightarrow \lambda y$$

Then:

$$f(\lambda x, \lambda y) = (\lambda x)^2 + (\lambda x)(\lambda y) + (\lambda y)^2$$

$$= \lambda^2(x^2 + xy + y^2)$$

Hence:

$$f(\lambda x, \lambda y) = \lambda^2 f(x, y)$$

Therefore:

$$f(x, y) \text{ is homogeneous of degree } 2$$

11.5 Euler's Theorem

Euler's Theorem

If:

$$z = f(x, y)$$

is homogeneous of degree n , then:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$$

Example

Verify Euler's theorem for:

$$z = x^2 + xy + y^2$$

Solution

$$\frac{\partial z}{\partial x} = 2x + y$$

$$\frac{\partial z}{\partial y} = x + 2y$$

Now:

$$\begin{aligned} & x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \\ &= x(2x + y) + y(x + 2y) \\ &= 2x^2 + xy + xy + 2y^2 \\ &= 2(x^2 + xy + y^2) \\ &= 2z \end{aligned}$$

Hence verified.

11.6 Maxima and Minima

Differentiation helps determine maximum and minimum values of functions.

11.6.1 Functions of One Variable

Suppose:

$$y = f(x)$$

Steps:

1. Find:

$$\frac{dy}{dx}$$

2. Solve:

$$\frac{dy}{dx} = 0$$

3. Find:

$$\frac{d^2y}{dx^2}$$

4. Apply second derivative test.

Example

Find maxima or minima of:

$$y = x^2 - 4x + 3$$

Solution

$$\frac{dy}{dx} = 2x - 4$$

Set equal to zero:

$$2x - 4 = 0$$

$$x = 2$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2$$

Since:

$$2 > 0$$

the function has minimum value.

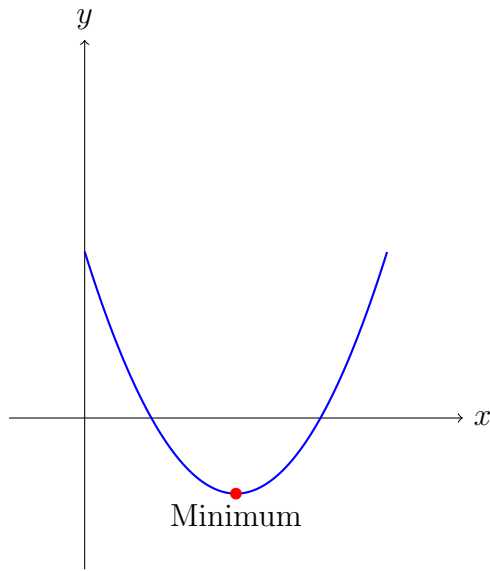
Value at:

$$x = 2$$

is:

$$y = 4 - 8 + 3 = -1$$

Minimum Value = -1



11.6.2 Functions of Two Variables

Suppose:

$$z = f(x, y)$$

For stationary points:

$$\frac{\partial z}{\partial x} = 0$$

and

$$\frac{\partial z}{\partial y} = 0$$

Example

Find stationary point of:

$$z = x^2 + y^2 - 4x - 6y$$

Solution

$$\frac{\partial z}{\partial x} = 2x - 4$$

$$\frac{\partial z}{\partial y} = 2y - 6$$

Set equal to zero:

$$2x - 4 = 0 \Rightarrow x = 2$$

$$2y - 6 = 0 \Rightarrow y = 3$$

Stationary point:

$$(2, 3)$$

Value:

$$z(2, 3) = 4 + 9 - 8 - 18$$

$$= -13$$

Minimum Value = -13

Important Notes

- Partial derivatives are used for multivariable functions.
- Homogeneous functions satisfy scaling property.
- Euler's theorem applies to homogeneous functions.
- Maxima and minima are useful in optimization problems.
- Partial differentiation is important in Economics, Physics and Engineering.

11.7 Solved Questions on Maxima and Minima

Question 1

Find the maximum or minimum value of:

$$y = x^2 - 6x + 5$$

Solution

First derivative:

$$\frac{dy}{dx} = 2x - 6$$

Set equal to zero:

$$2x - 6 = 0$$

$$x = 3$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2$$

Since:

$$2 > 0$$

the function has a minimum point.

Now find the value:

$$y(3) = 9 - 18 + 5$$

$$= -4$$

Minimum Value = -4

Question 2

Find the maximum or minimum value of:

$$y = 4x - x^2$$

Solution

Differentiate:

$$\frac{dy}{dx} = 4 - 2x$$

Set equal to zero:

$$4 - 2x = 0$$

$$x = 2$$

Second derivative:

$$\frac{d^2y}{dx^2} = -2$$

Since:

$$-2 < 0$$

the function has a maximum point.

Value at:

$$x = 2$$

$$y = 4(2) - 2^2$$

$$= 8 - 4$$

$$= 4$$

Maximum Value = 4

Question 3

Find maxima or minima of:

$$y = x^3 - 3x^2 + 2$$

Solution

First derivative:

$$\frac{dy}{dx} = 3x^2 - 6x$$

$$= 3x(x - 2)$$

Critical points:

$$x = 0, \quad x = 2$$

Second derivative:

$$\frac{d^2y}{dx^2} = 6x - 6$$

At:

$$x = 0$$

$$\frac{d^2y}{dx^2} = -6 < 0$$

Hence maximum point.

$$y(0) = 2$$

At:

$$x = 2$$

$$\frac{d^2y}{dx^2} = 6 > 0$$

Hence minimum point.

$$y(2) = 8 - 12 + 2$$

$$= -2$$

Therefore:

$$\boxed{\text{Maximum Value} = 2}$$

$$\boxed{\text{Minimum Value} = -2}$$

Question 4

Find the stationary point of:

$$y = x^2 + 8x + 7$$

Solution

Differentiate:

$$\frac{dy}{dx} = 2x + 8$$

Set equal to zero:

$$2x + 8 = 0$$

$$x = -4$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2$$

Since:

$$2 > 0$$

minimum point exists.

Now:

$$y(-4) = 16 - 32 + 7$$

$$= -9$$

$$\boxed{\text{Minimum Value} = -9}$$

Question 5

Find the maxima or minima of:

$$y = x^4 - 4x^2$$

Solution

First derivative:

$$\begin{aligned}\frac{dy}{dx} &= 4x^3 - 8x \\ &= 4x(x^2 - 2)\end{aligned}$$

Critical points:

$$x = 0, \quad x = \pm\sqrt{2}$$

Second derivative:

$$\frac{d^2y}{dx^2} = 12x^2 - 8$$

At:

$$x = 0$$

$$\frac{d^2y}{dx^2} = -8 < 0$$

Hence maximum point.

$$y(0) = 0$$

At:

$$x = \pm\sqrt{2}$$

$$\frac{d^2y}{dx^2} = 16 > 0$$

Hence minimum point.

Now:

$$\begin{aligned}y(\pm\sqrt{2}) &= (\sqrt{2})^4 - 4(\sqrt{2})^2 \\ &= 4 - 8 \\ &= -4\end{aligned}$$

Therefore:

Maximum Value = 0

Minimum Value = -4

11.8 Practice Questions

1. Find:

$$\frac{\partial z}{\partial x}$$

and

$$\frac{\partial z}{\partial y}$$

for:

$$z = x^2y + xy^2$$

2. Show that:

$$x^2 + y^2 + 2xy$$

is homogeneous.

3. Verify Euler's theorem for:

$$z = x^2 + 2xy + y^2$$

4. Find maxima or minima of:

$$y = x^2 - 6x + 5$$

5. Find stationary point of:

$$z = x^2 + y^2 - 2x - 4y$$